

HAOMING WANG

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RESEARCH INTERESTS

On-Device Multimodal AI, Vision-Language Models, Visual Spatial Reasoning, Generative-Model Interpretability, Federated Learning, VLM-Based Autonomous Mobile Systems.

EDUCATION

Sept 2022 – May 2027 (Anticipated)	Ph.D. in Electrical and Computer Engineering University of Pittsburgh, Pittsburgh, PA, USA Advisor: Prof. Wei Gao, Intelligent System Lab	
Sept 2018 – May 2022	B.Eng. in Automation Zhejiang University, Hangzhou, China Chu Kochen Honors College	GPA: 3.8/4.0

EXPERIENCE

Sept 2022 – Present	Graduate Student Researcher Intelligent System Lab, University of Pittsburgh, Pittsburgh, PA, USA • (May 2025 – Present) Led research on on-device VLM spatial reasoning: photo-realistic 3D benchmark synthesis [1]; on-device cross-frame reasoning deployed on Jetson Orin / Meta AR Glass / smartphone with full memory and latency profiling [5]; Dirichlet-energy latent shaping of VLM 3D-topology subspaces [6]. • (Nov 2024 – May 2025) Drove research on on-device LLM personalization via explainable model selection [2], and training-free fine-grained interpretability for diffusion / GAN / autoregressive image generators [7]. • (Sept 2022 – Oct 2024) Designed federated learning systems for heterogeneous AIoT fleets: gradient correction under intertwined data / device heterogeneity [3]; federated learning with unbounded device staleness via gradient inversion [4].
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PUBLICATIONS (PEER-REVIEWED)

1. **[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
Haoming Wang, Qiyao Xue, Wei Gao.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (Acceptance 25.4%, Oral).
[\[arXiv\]](#) [\[Code\]](#) [\[Data\]](#)
 - **Architected** a customizable framework for analyzing VLM spatial reasoning that generates an unlimited supply of photo-realistic 3D test scenes with user-controllable axes for object count, clutter, occlusion, and viewpoint.
2. **[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection**
Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao.
Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025 (Acceptance 18.0%). [\[Paper\]](#)
 - **Developed Xpert**, an explainable model-selection method that warm-starts on-device LLM personalization from the closest pre-personalized checkpoint of a similar past user, eliminating cold-start cost on edge hardware.
3. **[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications**

Haoming Wang, Wei Gao.

Proceedings of the 31st ACM International Conference on Mobile Computing and Networking (MobiCom), 2025 (Acceptance 17.1%). [\[Paper\]](#)

- **Engineered** a federated AIoT training method that jointly disentangles and corrects for data heterogeneity and device delays via targeted gradient correction, restoring clean convergence on heterogeneous device fleets.

4. **[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness**

Haoming Wang, Wei Gao.

Proceedings of the 39th AAAI Conference on Artificial Intelligence, 2025 (Acceptance 23.4%). [\[arXiv\]](#)

- **Designed** a gradient-inversion technique that recovers usable training signal from arbitrarily stale device updates, enabling FL to converge under the unbounded staleness regime real mobile / IoT fleets operate in.

PREPRINTS & PAPERS UNDER REVIEW

5. **MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation**

Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao.

Under Review, 2026. [\[arXiv\]](#)

- **Built** a training-free pipeline that stitches multi-view RGB frames into a compact semantic map, boosting small on-device VLMs' cross-frame spatial reasoning by up to 40%; deployed on Jetson Orin, Meta AR Glass, and smartphone.

6. **Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models**

Haoming Wang, Wei Gao.

Preprint, 2026. [\[Paper\]](#) [\[Code\]](#)

- **Discovered** a VLM latent subspace encoding 3D scene topology and **introduced** a Dirichlet-energy regularizer that improves real-world spatial reasoning by up to 12.1% over standard SFT.

7. **Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL)**

Haoming Wang, Wei Gao.

Under Review, 2025. [\[Paper\]](#)

- **Architected FineXL**, a training-free method that decomposes a personalized image generator's divergence from its base into orthogonal natural-language concepts; cut single-aspect explanation error by 56% across diffusion, GAN, and autoregressive generators.

8. **FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing**

Kai Huang, **Haoming Wang** (co-author), Wei Gao.

Preprint, 2024. [\[Paper\]](#) [\[Code\]](#)

- **Co-developed** a bilevel-optimization defense that learns which diffusion-model tensors to lock so users cannot fine-tune toward illegal content; 37–47% stronger mitigation than unlearning baselines.

9. **HEVEN: Small Object Search and Navigation on VLM-Based Autonomous Mobile Systems**

Anonymous Authors (incl. **Haoming Wang**, 3rd author).

Double-Blind Under Review, 2026.

- **Co-architected** the first VLM-based mobile system for small-object search, with a hierarchical spatial semantic tree and Tree-of-Thought planner; achieved 73.86% success rate (3× over SOTA) on a real quadruped robot.

10. **GlobalNav: Daily Object Navigation in VLM-Based Autonomous Mobile Systems with Aligned Local and Global Views**

Anonymous Authors (incl. **Haoming Wang**, 3rd author).

Double-Blind Under Review, 2026.

- **Co-developed** an infrastructure-assisted VLM navigation system aligning ceiling-camera global views with the robot's first-person view; lifted daily-object navigation success from <30% to 90.91% in simulation.

11. **Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective**

Qiyao Xue, **Haoming Wang** (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao.

Under Review, 2026. [\[arXiv\]](#)

- **Co-authored** ReMindView-Bench, a cognitively grounded multi-view spatial-reasoning benchmark (50K+ VQAs across 15 VLMs) with a joint reasoning-path / latent-state analysis pipeline.

12. **Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods**

Weichen Liu, Qiyao Xue, **Haoming Wang**, Xiangyu Yin, Boyuan Yang, Wei Gao.

Under Review, 2026.

HONORS & AWARDS

April 2026

ECE Research Assistant of the Year

Department of Electrical and Computer Engineering, University of Pittsburgh

TECHNICAL SKILLS

On-Device / Edge: Jetson Orin (PyTorch + TensorRT), Meta AR Glass, Android (ONNX / NNAPI), Unitree Go2, RealSense D435i; memory / latency profiling across 3B–32B VLMs.

Models & Tools: Stable Diffusion, CLIP, Gemini, GPT-4o/5, InternVL, Qwen-VL, LLaVA-Video, Grounding-SAM, DepthPro.

Programming: Python, PyTorch, CUDA, C/C++.